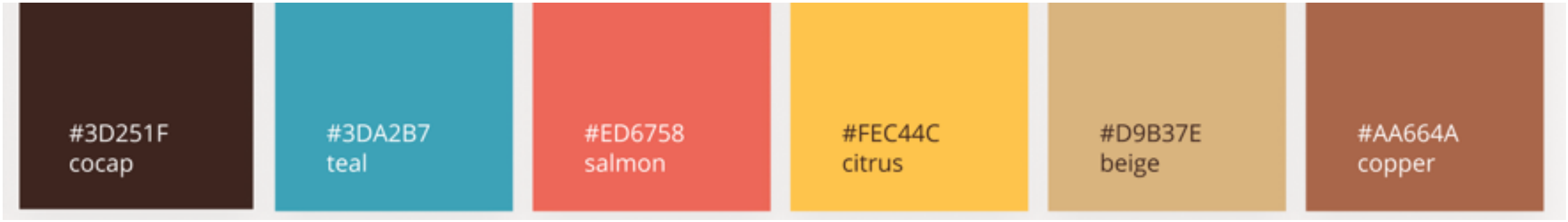


Large n (e.g. images, text, biology) → intractable

$\pi^\star$ is often sparse and unstable

Makes optimization strictly convex.



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Motivations for Regularization

- Computational challenges:
 - Discrete OT = Linear Program with $\mathcal{O}(n^2)$ variables.
 - Worst-case complexity: cubic or worse (on $N = n^2$ variables!).
 - **Large n (e.g. images, text, biology) → intractable**
- Practical issue:
 - Optimal plan $\pi^\star$ **is often sparse and unstable** (recall Kantorovich $\Leftrightarrow$ Monge in certain settings!)
 - Small data perturbations $\rightarrow$ large changes in $\pi^\star$.
- Idea: add an entropy term to smooth solutions.
 - Encourages more diffuse transport plans.
 - **Makes optimization strictly convex.**

Entropy of π